

1 Solution — GIAM 3.4

Problem 5

1.1 Problem

True or false: the sum of any two irrational numbers is irrational. Prove your answer.

1.2 Answer

False. $\sqrt{2} + (-\sqrt{2}) = 0.$

1.3 Disproof

The claim is universally quantified, so its negation is existential and one instance settles it:

$$\neg(\forall x, y \in \mathbb{R} \setminus \mathbb{Q}, x + y \notin \mathbb{Q}) \cong \exists x, y \in \mathbb{R} \setminus \mathbb{Q}, x + y \in \mathbb{Q}.$$

Disproof. Take $x = \sqrt{2}$ and $y = -\sqrt{2}$.

$\sqrt{2}$ is irrational (Theorem 1.6 of the text). And $-\sqrt{2}$ is irrational too: if $-\sqrt{2} = q$ for some rational q , then $\sqrt{2} = -q$, and $-q$ is rational since $\mathbb{Q}$ is closed under negation — contradicting the irrationality of $\sqrt{2}$.

So x and y are both irrational, yet

$$x + y = \sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}.$$

The hypothesis of the claim is satisfied and its conclusion fails, so the claim is false. ■

It is worth stressing that the second sentence is not padding. A counterexample only works if you check that it *is* one, and “ $-\sqrt{2}$ is irrational” needs the one-line argument above rather than being assumed.

If the minus sign feels like a trick, an equally valid pair of **positive** irrationals works:

$$\left(2-\sqrt{2}\right)+\sqrt{2}=2,\qquad\text{with }2-\sqrt{2}\approx 0.5858.$$

Here $2-\sqrt{2}$ is irrational by 3.3 Problem 5 — a rational plus an irrational is irrational.

1.4 The Picture

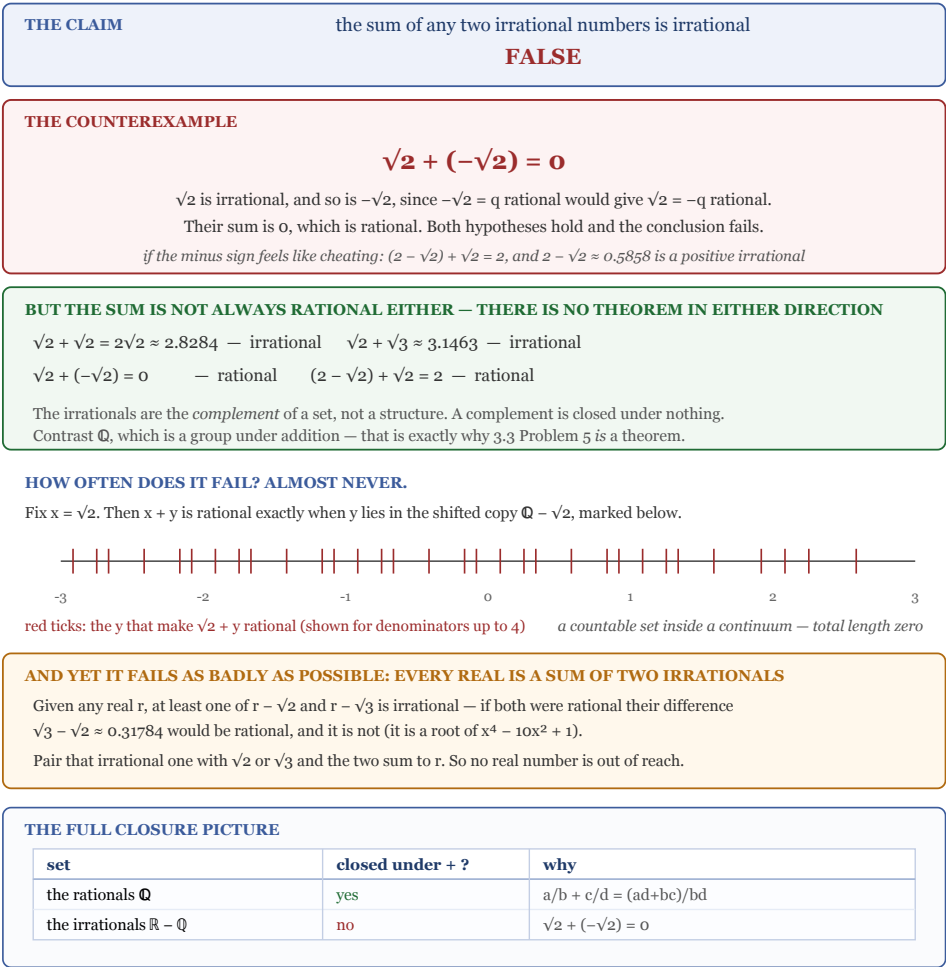


Figure 1.1: A diagram in six bands. The first states the claim — the sum of any two irrational numbers is irrational — and the verdict, FALSE. The

second gives the counterexample: root 2 plus minus root 2 equals 0. Root 2 is irrational and so is minus root 2, since minus root 2 equal to a rational q would give root 2 equal to minus q , also rational. Their sum is 0, which is rational; both hypotheses hold and the conclusion fails. If the minus sign feels like cheating, 2 minus root 2 plus root 2 equals 2, and 2 minus root 2, about 0.5858, is a positive irrational. The third band notes that the sum is not always rational either, so there is no theorem in either direction: root 2 plus root 2 is about 2.8284 and irrational, root 2 plus root 3 is about 3.1463 and irrational, while root 2 plus minus root 2 is 0 and 2 minus root 2 plus root 2 is 2, both rational. The irrationals are the complement of a set, not a structure, and a complement is closed under nothing; contrast the rationals, which form a group under addition, which is exactly why 3.3 Problem 5 is a theorem. The fourth band asks how often it fails and answers: almost never. Fixing x equal to root 2, the sum x plus y is rational exactly when y lies in the shifted copy of the rationals minus root 2, drawn as a row of sparse red ticks along a number line from minus 3 to 3 for denominators up to 4 — a countable set inside a continuum, of total length zero. The fifth band says that nonetheless it fails as badly as possible, since every real is a sum of two irrationals: given any real r , at least one of r minus root 2 and r minus root 3 is irrational, because if both were rational their difference root 3 minus root 2, about 0.31784, would be rational, and it is not, being a root of x to the fourth minus 10 x squared plus 1. Pairing that irrational one with root 2 or root 3 gives two irrationals summing to r , so no real number is out of reach. The final band tabulates closure: the rationals are closed under addition because a over b plus c over d equals ad plus bc over bd , while the irrationals are not, since root 2 plus minus root 2 equals 0.

1.5 Remark — There Is No Theorem in Either Direction

A natural follow-up is whether the claim is at least *reversible* — perhaps the sum of two irrationals is always rational? It is not:

$\sqrt{2} + \sqrt{2} = 2\sqrt{2} \approx 2.8284$	irrational
$\sqrt{2} + \sqrt{3} \approx 3.1463$	irrational
$\sqrt{2} + (-\sqrt{2}) = 0$	rational
$(2 - \sqrt{2}) + \sqrt{2} = 2$	rational

So the sum of two irrationals can be anything at all, and no implication holds either way. That is the honest state of affairs, and it is worth saying explicitly: disproving a claim does not automatically establish its opposite.

The structural reason is that the irrationals are defined as a **complement**, $\mathbb{R} \setminus \mathbb{Q}$, rather than by any closure property. A complement inherits no algebra. Compare 3.3 Problem 5, where *rational plus irrational is irrational* really is a theorem — precisely because $\mathbb{Q}$ is a group under addition, so subtracting a rational cannot escape it.

1.6 Remark — Why the Claim Feels True: It Almost Always Is

The conjecture is not stupid, and it is worth seeing why. Fix an irrational x . Then

$$x + y \in \mathbb{Q} \iff y \in \mathbb{Q} - x = \{q - x : q \in \mathbb{Q}\},$$

which is a shifted copy of $\mathbb{Q}$ — a **countable** set. So among the uncountably many irrational y , only countably many produce a rational sum. The failures have Lebesgue measure zero.

Pick two irrationals “at random” in any reasonable sense and the sum is irrational with probability 1. The claim is true almost everywhere and false nonetheless, which is exactly the situation Section 3.4 exists to warn about: overwhelming statistical support is not proof, and a measure-zero exception refutes a universal statement just as decisively as a common one.

1.7 Remark — How Badly It Fails: Every Real Is a Sum of Two Irrationals

Although failures are rare for a *fixed* x , if both summands are allowed to vary the claim collapses completely.

Claim. *Every real number r is the sum of two irrational numbers.*

Proof. *Consider $r - \sqrt{2}$ and $r - \sqrt{3}$. At least one is irrational: if both were rational, their difference*

$$(r - \sqrt{2}) - (r - \sqrt{3}) = \sqrt{3} - \sqrt{2}$$

would be rational, and it is not — $\sqrt{3} - \sqrt{2} \approx 0.31784$ is a root of $x^4 - 10x^2 + 1$, which the rational root theorem shows has no rational roots. If $r - \sqrt{2}$ is irrational, write $r = (r - \sqrt{2}) + \sqrt{2}$; otherwise write $r = (r - \sqrt{3}) + \sqrt{3}$. Either way r is a sum of two irrationals. ■

So the set of achievable sums is not merely “sometimes rational” — it is all of $\mathbb{R}$. Nothing whatsoever can be concluded about $x + y$ from the irrationality of x and y .

1.8 Remark — What Survives, and Where the Line Falls

Gathering the results of this problem and its neighbours gives a complete picture of which combinations are theorems:

statement	true?	witness or reason
rational + rational is rational	yes	$\mathbb{Q}$ closed under +
rational + irrational is irrational	yes	3.3 Problem 5
irrational + irrational is irrational	no	$\sqrt{2} + (-\sqrt{2}) = 0$
irrational $\times$ irrational is irrational	no	$\sqrt{2} \cdot \sqrt{2} = 2$

The pattern is that a statement survives exactly when one operand is drawn from the *structured* set. Mixing a group element with a non-member is controlled; combining two non-members is not, because the two “irrational parts” are free to cancel.

Seen through the coset language of 3.3 Problem 5: adding a rational moves you *along* a coset of $\mathbb{Q}$, which never returns you to $\mathbb{Q}$ itself; adding a second irrational moves you to a *different* coset, and one of the available destinations is $\mathbb{Q}$. Cancellation is not an accident of $\sqrt{2}$ — it is built into the group structure, since x and $-x$ always lie in cosets that sum to the identity one.